PN junctions in semiconductors

Part A. Electrostatics (5.0 points)

A.1 (0.5pt)
Relation:

A.2 (0.5pt)
Proof:

A.3 (1.0pt)

$E_x(x) =$

A.4 (1.0pt)

$\varphi(x) =$

A.5 (0.5pt)
Plot:

A.6 (1.0pt)

(Formula) $d_n =$ (Value) $d_n =$

(Formula) $d_p =$ (Value) $d_p =$

A.7 (0.5pt)

(Formula) $E_{\max} =$ (Value) $E_{\max} =$

Part B. Junction with external voltage (5.0 points)

B.1 (0.5pt)

$n_1 =$

$p_1 =$

B.2 (1.0pt)
Equation:

B.3 (0.6pt)

(Formula) $L_p =$ (Value) $L_p =$

B.4 (0.8pt)

$p_1(x) =$

B.5 (0.8pt)

$j_p =$

B.6 (0.8pt)

$I =$

B.7 (0.3pt)
Plot:

B.8 (0.2pt)

$I_\infty =$

$I_1 =$